

Human-in-the-Loop Framework for **Text-to-Tone** Generation and Refinement

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Question Statement

Abstract

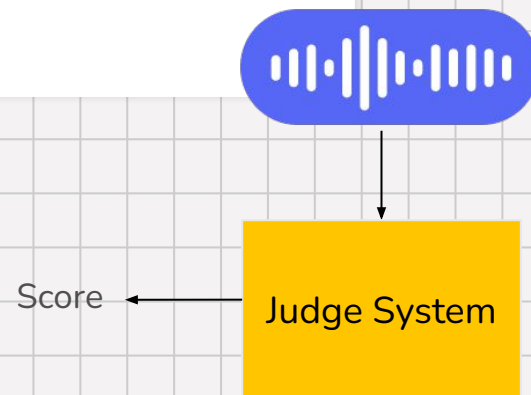
- Nowadays, LLMs and generative audio models can already produce tones/music from text. (*Can Large Language Models Predict Audio Effects Parameters from Natural Language?*)
- But there is still no reliable way to **evaluate** whether those tones actually match user intention or its quality.

Related Works

- 1) **Embedding-based** evaluation (*CLAP, MERT with MAD, FAD*)
 - > Only measure similarity, not quality
 - > *CLAP*: general for audio to text
 - > *MERT*: musical embedding to represent emotional / textual alignment
- 2) **Optimization-based** methods (*Text2FX*)
 - > Can't evaluate output (listening test)
 - > Can't refine parameters based on human feedback

Research Abstract

- Create an fair **evaluation system** for tone/music
- **Refine** mechanism for editing the generated tone/music
(re-prompting, human-in-the-loop...)

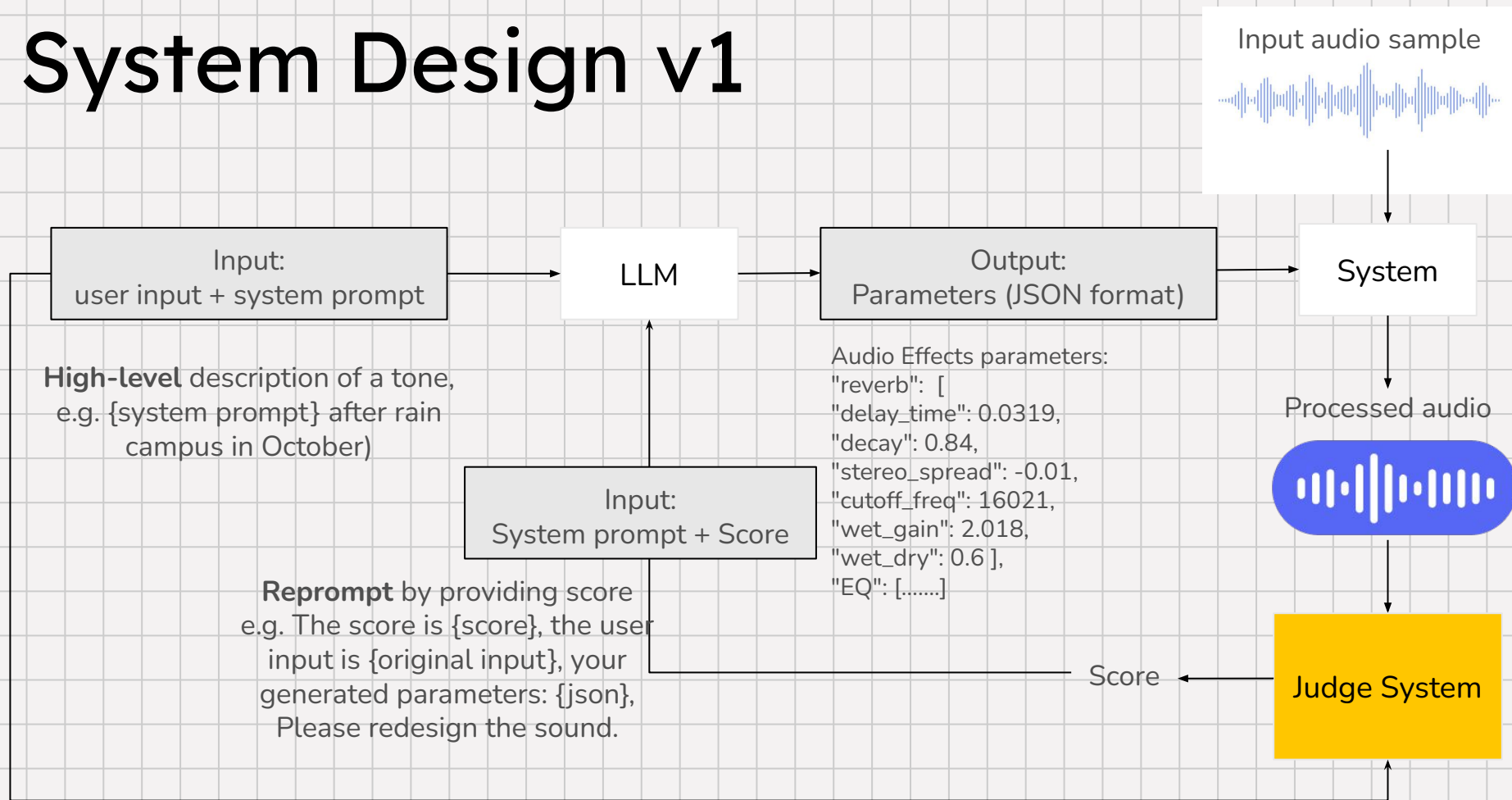


Experiment Method v.1



A **closed-loop** architecture enabling LLMs to iteratively **refine tone design** based on **self-evaluated scores**.

System Design v1



Pipeline Demo

This demo will be our testing baseline for our experiment.

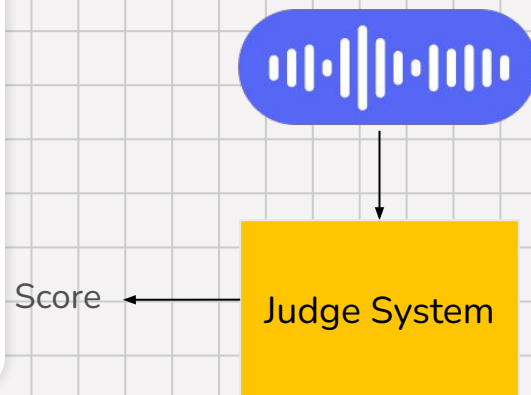
Judge System - *Our Main Contribution*



Each score = combination of **quantitative** and **qualitative** components:

Type	Source	Description
AudioScore	Multimodal embeddings similarity	Top-k labels of the processed audio
ToneJudge	LLM-as-judge	User input + output json + audio extracted features (MFTT, spectrum...)
NIMA-liked	User real data	Human-in-the-loop

$$R = \alpha \cdot R_{\text{ToneJudge}} + \beta \cdot R_{\text{AudioScore}} + \gamma \cdot R_{\text{NIMA}}$$

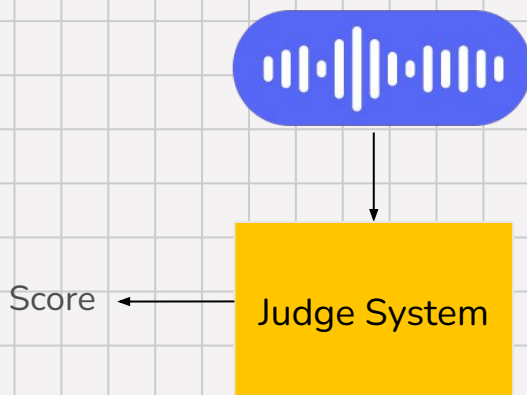


Dataset & Experiments (v1)

Dataset: Used **SocialFX** dataset

Method:

- Top-k CLAP similarity
- Top-k MERT + CLAP???

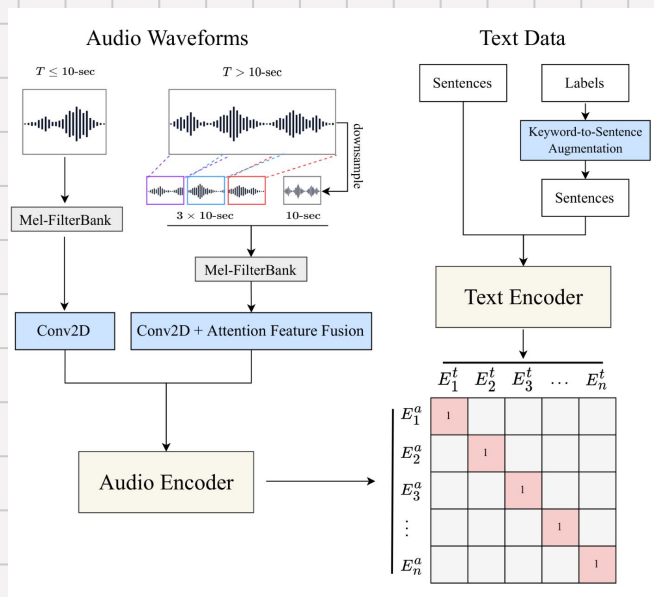


Split (3)
reverb · 6.77k rows

Search this dataset

id string · lengths	text string · lengths	param_values list · lengths	param_keys list · lengths	extra string · lengths
8	11	5	6	54
reverb_0	live-concert	[0.0319, 0.84, -0.01,...	["delay_time"...	{'lang': 'English', 'agreed': 'electric', 'disnotagree':...
reverb_1	fantasy	[0.0559, 0.94,...	["delay_time"...	{'lang': 'English', 'agreed': 'warm', 'disnotagree': 'big-...
reverb_2	dizzy	[0.0319, 0.96,...	["delay_time"...	{'lang': 'English', 'agreed': 'unpleasant,e much,echoing,distortion', 'disnotagree':...
reverb_3	echoing,chrystal,angels,heaven, harp-strums	[0.01, 0.99, -0.0043,...	["delay_time"...	{'lang': 'English', 'agreed': 'spacious', 'disnotagree': 'clashing,muted,clash,awesom
reverb_4	empty,spacious,softer	[0.0188, 0.7, 0.0035,...	["delay_time"...	{'lang': 'English', 'agreed': nan, 'disnota nan}
reverb_5	richer,exaggerated,dramatic	[0.0559, 0.94,...	["delay_time"...	{'lang': 'English', 'agreed': 'spacious', 'disnotagree': 'electric,echoes,soft,distan
reverb_6	spacious,hall,echo,vibrate	[0.0329, 0.8, 0.0044,...	["delay_time"...	{'lang': 'English', 'agreed': 'spacey,big', 'disnotagree':...
reverb_7	treble	[0.0879, 0.9, -0.0096...	["delay_time"...	{'lang': 'English', 'agreed': 'extended', 'disnotagree': 'horn-...
reverb_8	mixed,multiple-effect	[0.0109, 0.01,...	["delay_time"...	{'lang': 'English', 'agreed': 'increased- base,sharp', 'disnotagree':...
reverb_9	humongous	[0.0568, 0.91,...	["delay_time"...	{'lang': 'English', 'agreed': 'big- room,hollow,cavernous,large,big', 'disnota
reverb_10	louder,bolder	[0.0879, 0.9, -0.0112...	["delay_time"...	{'lang': 'English', 'agreed': 'louder,loud', 'disnotagree':...
reverb_11	large,church-bells	[0.0319, 0.96...	["delay_time"...	{'lang': 'English', 'agreed': 'melody', 'disnotagree': 'live.rock.too-...

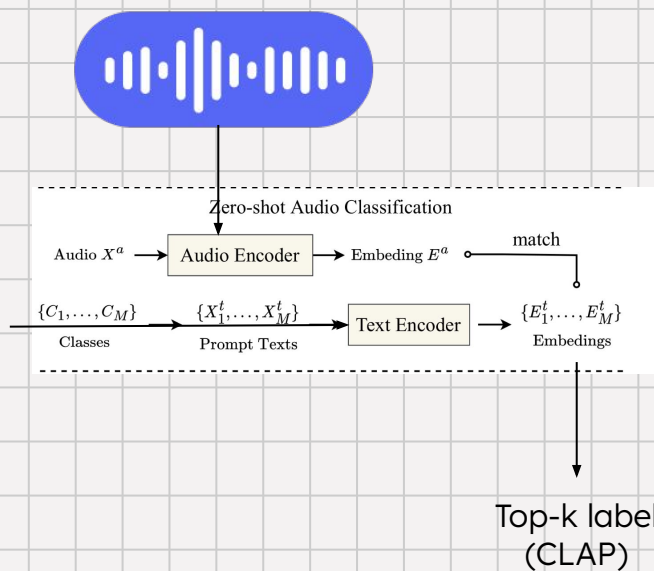
1. Top-k CLAP



Label from SocialFX

```
baseline-system > data > cleaned_labels > {} reverb_labels.json
1
2 "effect_type": "reverb",
3 "labels": [
4   "a-echo-howling",
5   "a-nice",
6   "abrasive",
7   "acoustic",
8   "acoustical",
9   "action-invoking",
10  "active",
11  "affective",
12  "agitated",
13  "agony",
14  "air-like",
15  "airy",
16  "alarming",
17  "alert",
18  "alien",
19  "alive",
20  "all-around",
21  "alone",
22  "altering",
23  "alternative-rock",
24  "amazing",
25  "ambience",
26  "ambient",
27  "ambient-trails",
28  "amplified",
29  "ancient",
30  "angelic",
31  "angry",
32  "annoying",
33  "antique",
34  "appealing",
35  "approaching-stormily",
36  "arena",
37  "army",
38  "atmospheric",
```

Processed Audio (dry + ground truth effects)



2. Top-k MERT

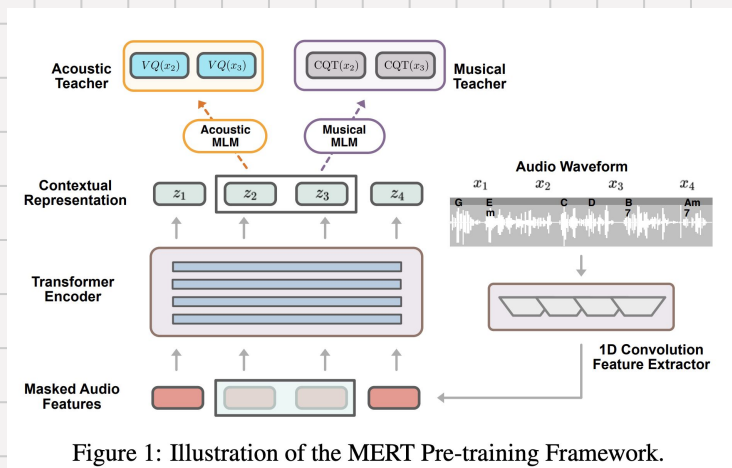
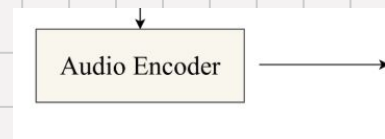


Figure 1: Illustration of the MERT Pre-training Framework.

Processed Audio
(dry + ground truth effects)



Audio embeddings
(MERT)



Audio embeddings
(CLAP)

Top-k label
(MERT)

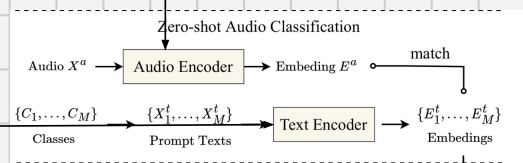


3. Hybrid

Label from SocialFX

```
baseline-system > data > cleaned_labels > {} reverb_labels.json
1
2 "effect_type": "reverb",
3 "labels": [
4   "a-echo-howling",
5   "a-nice",
6   "abrasive",
7   "acoustic",
8   "acoustical",
9   "action-invoking",
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17  "alert",
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31  "angry",
32  "annoying",
33  "antique",
34  "appealing",
35  "approaching-stormily",
36  "arena",
37  "army",
38  "atmospheric",
```

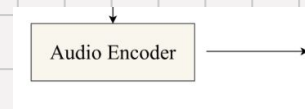
Processed Audio
(dry + ground truth effects)



a * Top-k label
(CLAP)

+ **b** * Top-k label
(MERT)

Processed Audio
(dry + ground truth effects)



Audio embeddings
(MERT)

Linear

Audio embeddings
(CLAP)

Case study - spacious.wav



```
=====
SUMMARY COMPARISON
=====
```

Rank	CLAP	MERT	Hybrid
1	huge-trumpets	engulfing	classy
2	chaotic	classy	huge-trumpets
3	cathedral	run-together	antique
4	big-church	liquid	old-organ
5	cathedral-like	low-fidelity	dying
6	church-bell	roomy	dominant
7	glorious	wavy	low-fidelity
8	tunnel-like	high-bass	airy
9	old-timely	commanding	serene
10	ominous	serene	chambers

Case study - warm.wav



```
=====
SUMMARY COMPARISON
=====
```

Rank	CLAP	MERT	Hybrid
1	church-bell	high-bass	classy
2	big-church	classy	low-fidelity
3	cathedral	low-fidelity	high-bass
4	cathedral-like	engulfing	chambers
5	chaotic	roomy	old-organ
6	church-like	wavy	old-timely
7	glorious	high-pitch	waved
8	ominous	large-auditorium	dominant
9	chambers	liquid	mixed
10	classical	comfortable	loving

Case study - muffled.wav



=====

SUMMARY COMPARISON

=====

Rank	CLAP	MERT	Hybrid
1	meditative	high-bass	wavy
2	lovely-echoes	engulfing	lovely-echoes
3	gothic	classy	loving
4	majestic	wavy	spacious
5	reflections	robust	soothing
6	melody	eco	run-together
7	introspective	roomy	vintage
8	jazzy	spacious	nicer
9	glorious	clear	gothic
10	mystery	muted	lengthy

No HIT in top 10.....

REVERB Effect

Best Configuration: **Cleaned labels + MERT mode**

Cleaned + CLAP

837 labels | 11 samples

Hit@1:	0.000
Hit@5:	0.000
Hit@10:	0.000
MRR:	0.000

Original + CLAP

1890 labels | 11 samples

Hit@1:	0.000
Hit@5:	0.000
Hit@10:	0.000
MRR:	0.000

Cleaned + MERT

837 labels | 11 samples

Hit@1:	0.000
Hit@5:	0.091
Hit@10:	0.091
MRR:	0.018

Original + MERT

1890 labels | 11 samples

Hit@1:	0.000
Hit@5:	0.000
Hit@10:	0.091
MRR:	0.013

Cleaned + HYBRID

837 labels | 11 samples

Hit@1:	0.000
Hit@5:	0.000
Hit@10:	0.000
MRR:	0.000

Original + HYBRID

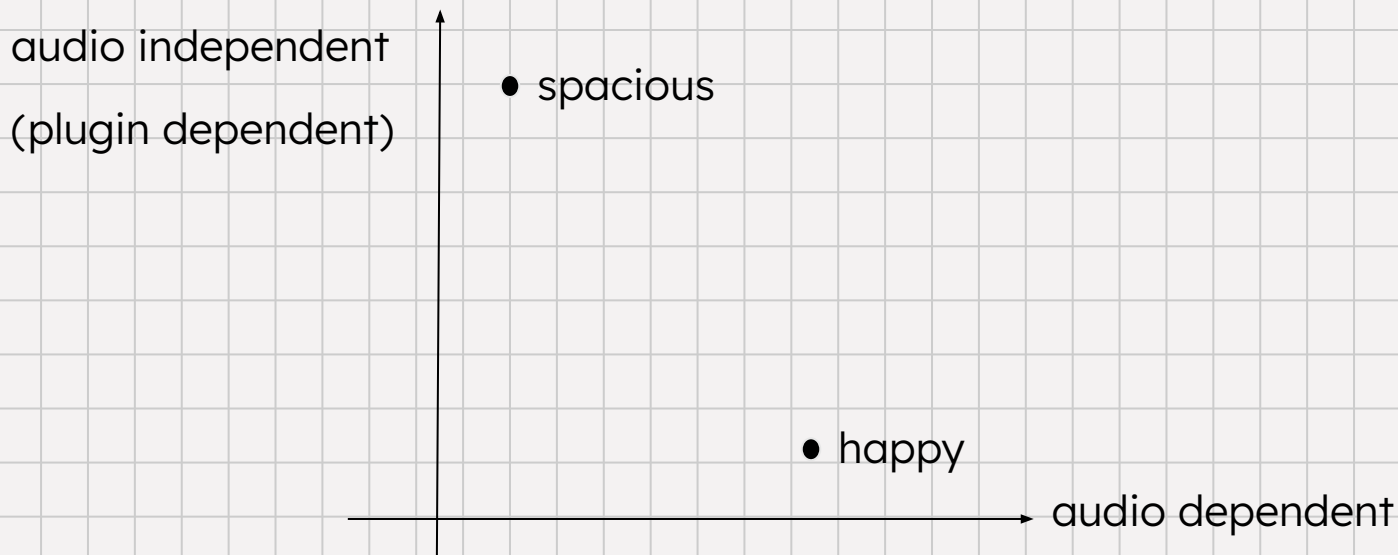
1890 labels | 11 samples

Hit@1:	0.000
Hit@5:	0.000
Hit@10:	0.000
MRR:	0.000

Hypothesis:

We assume the label can be separated into “**audio dependent**” & “audio independent”.

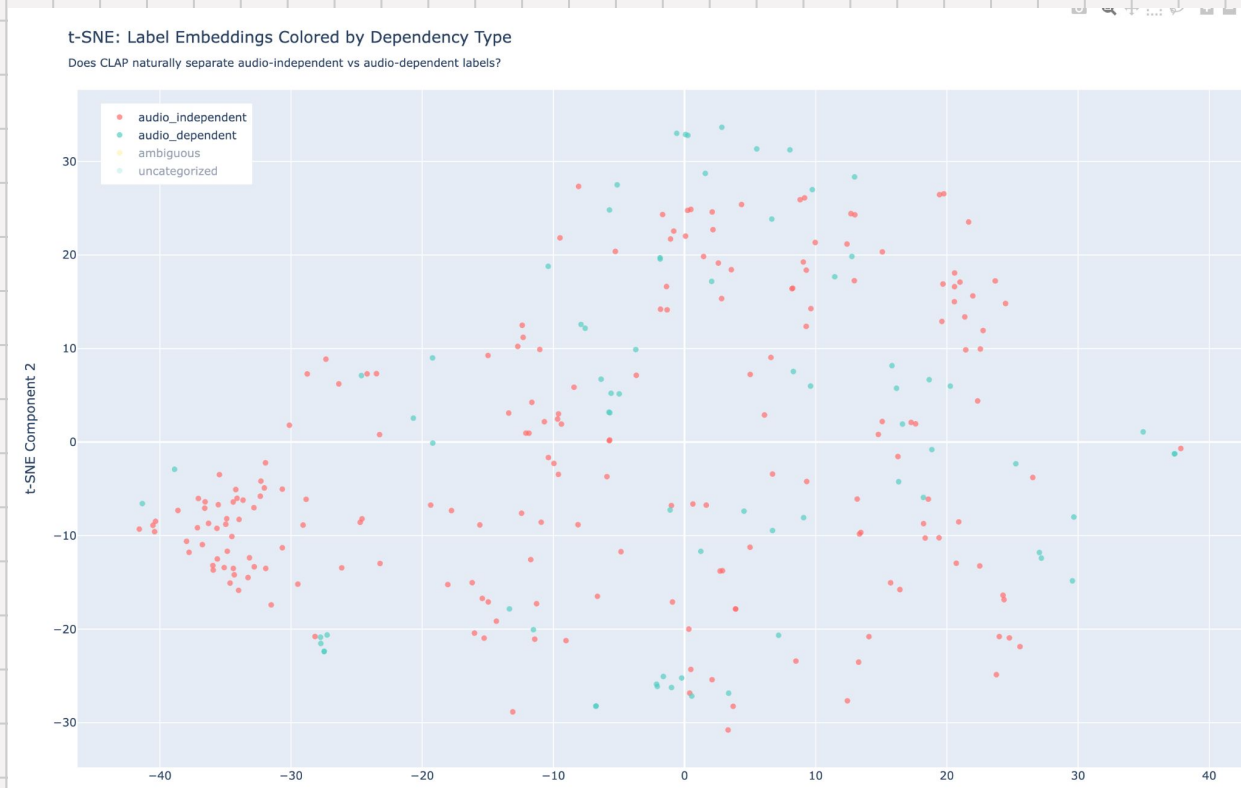
> Expect plugin-dependent words to be easier -> a better result in evaluation system.



Maybe deep into the data.....?

Tsne on CLAP? graphs?

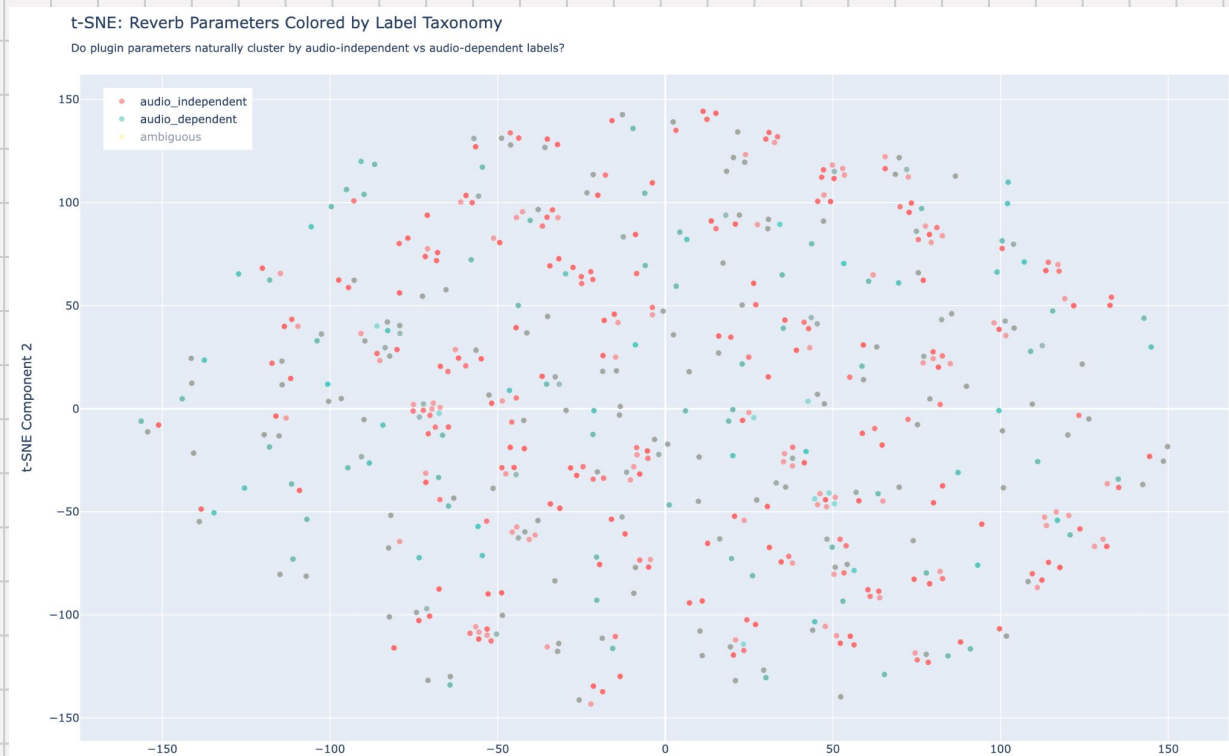
It's not separated in
CLAP text embeddings!



Maybe deep into the data.....?

Tsne on SocialFX? graphs?

descriptors are not
separable!



Problem & Challenges of v1

1. It's hard to provide a fair “score” in the judge system:

- *Semantic descriptors in SocialFX are extremely noisy.*
- *No standard answer for what “warm” or “airy” should be.*
- *How to quantize music into a score?*

Human-in-the-loop!

VibeCheck!

→ Reinforces that our judge system lacks semantic grounding.

2. Reprompting may NOT converge

Even if the judge outputs a score like 3/10 ($<$ threshold) and we reprompt the LLM:

“Given your score is 3/10, please redesign the sound.”

→ LLM reprompting is fundamentally unstable.

Optimization Problem!

Question Re-Statement

Instead of asking:

“How do we ***judge*** a sound?”

We should ask:

“How do we ***move*** a sound toward the user’s intention?”

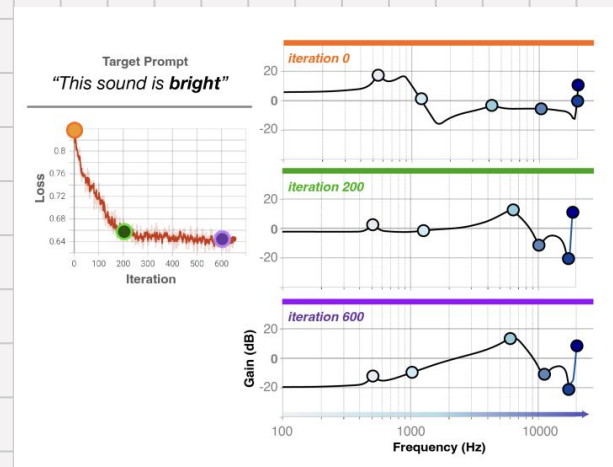
System Design v2: Human-in-the-Loop Optimization

Related Works

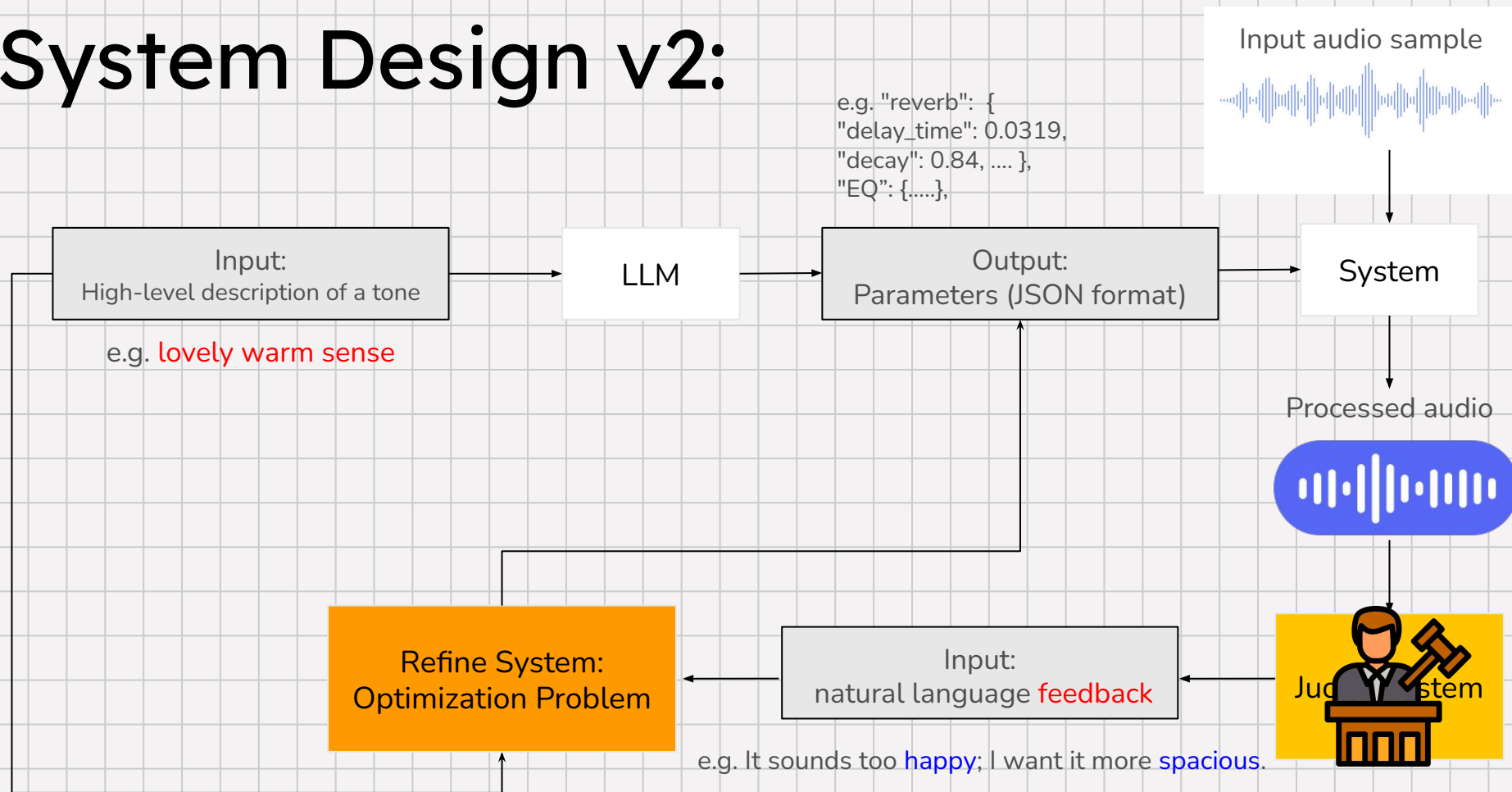
Inspired by *Text2FX*, *InstructPix2Pix*:

- *Text2FX* **optimizes** parameters toward a prompt
- *InstructPix2Pix* allows instruction-based refinement

We propose a **tone-editing system**, not just a tone generator.

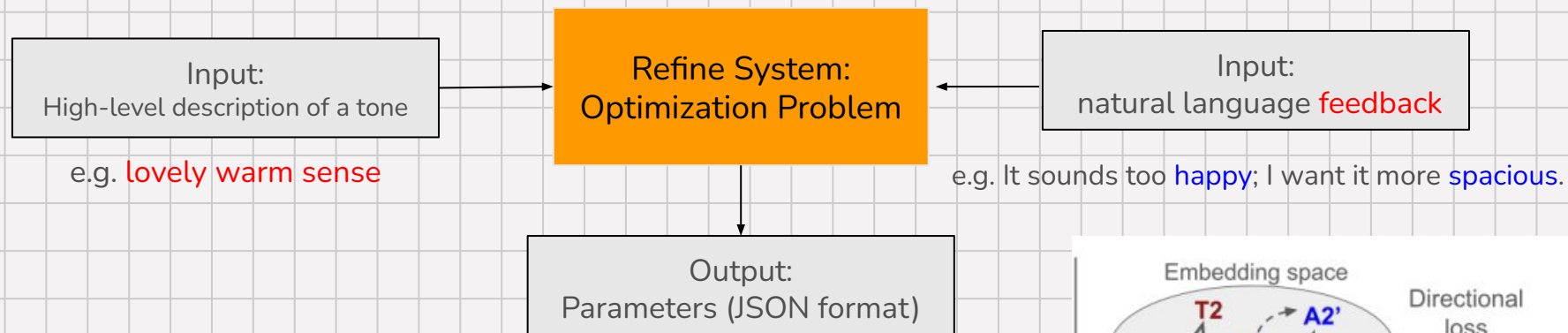


System Design v2:

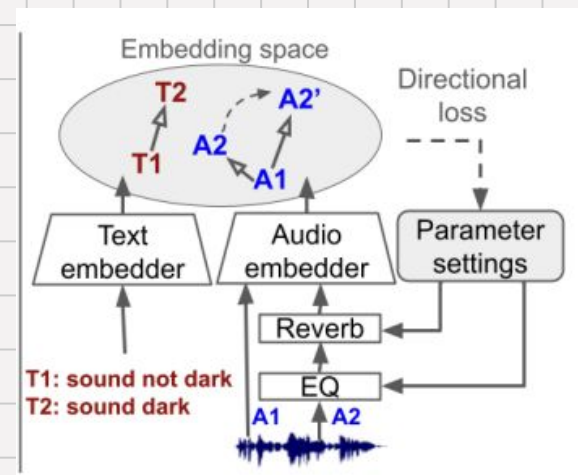


Refine System:

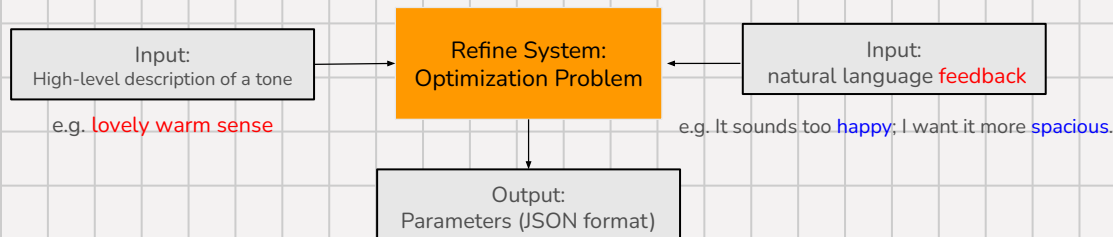
This avoids the need for a **hard judge** and solves the **convergence issue**.



1. Compute two semantic direction vectors:
 - Original prompt direction (→ lovely, warm)
 - User-feedback direction (→ not happy, → spacious)
2. Combine directions with learnable weights
3. Apply directional update to FX parameters



Experiments:



Baseline 1 – LLM reprompt

Expected: fails (no convergence)

Baseline 2 – Feedback directional update

Optimize audio embedding movement to match feedback text movement.

Baseline 3 – Hybrid weighted direction

Combined original Prompt direction + feedback direction combined.

Baseline 4 – Gene Algorithm / Simulated Annealing

Carmine's suggestion!

Baseline 5 – RL-based refinement

Q: What **dataset** we want to use?

A: Probably still SocialFX? Or even unsupervised?

Thank you!